

SECTION 1 Identification

1.1. Product identifier

Product form : Mixture
Trade name : Curator

1.2. Other means of identification

Other means of identification : EPA Reg. No.: 228-771

1.3. Recommended use of the chemical and restrictions on use

Use of the substance/mixture : Scientific research and development
Restrictions on use : All other uses not recommended above

1.4. Supplier's details

Nufarm Americas, Inc.
11901 South Austin Avenue
Alsip, IL 60803
United States
T 1-800-345-3330
www.nufarm.com

1.5. Emergency phone number

Emergency number : For Hazardous Materials or Dangerous Goods Incident Spill, Leak, Fire, Exposure, or Accident
Call CHEMTREC Day or Night: 1-800-424-9300 (Toll Free, USA) / 703-527-3887 (Virginia, USA)
CCN 16105
Back-up Emergency Number: +1 703-741-5970 (Washington, DC)

SECTION 2 Hazard Identification

2.1. Classification of the substance or mixture

GHS US classification

Flammable liquid, Category 4 H227 Combustible liquid.
Full text of H statements : see section 16

2.2. Label elements

GHS US labeling

Signal word (GHS US) : Warning
Hazard statements (GHS US) : H227 - Combustible liquid
Precautionary statements (GHS US) : Keep away from heat, hot surfaces, sparks, open flames, and other ignition sources. No smoking.
Wear protective gloves, protective clothing, eye and face protection.
Store in a well-ventilated place. Keep cool.
Dispose of contents and/or container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulations.

2.3. Hazards associated with known or reasonably anticipated uses

No additional information available

2.4. Hazards not otherwise classified

No additional information available

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According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

2.5. Unknown acute toxicity

No additional information available

SECTION 3 Composition/information on ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Name	Product identifier	%	GHS US classification
Propamacarb HCL*	CAS-No.: Trade Secret	40	Skin Irrit. 2, H315 Skin Sens. 1, H317
Propylene glycol	CAS-No.: 57-55-6	5 - 10	Not classified
Mefenoxam	CAS-No.: 70630-17-0	4	Acute Tox. 4 (Oral), H302 Eye Dam. 1, H318

*Chemical name, CAS number and/or exact concentration have been withheld as a trade secret

Full text of hazard classes and H-statements : see section 16

SECTION 4 First aid measures

4.1. Description of necessary first-aid measures

First-aid measures general	: First aider: Pay attention to self-protection. Give artificial respiration if necessary. Induce artificial respiration with mask fitted with one-way valve or other suitable device but, not mouth-to-mouth. Call a poison center/doctor/physician if you feel unwell.
First-aid measures after inhalation	: If breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing. If experiencing respiratory symptoms: Call a poison center or a doctor.
First-aid measures after skin contact	: Rinse skin with water/shower. If skin irritation occurs: Get medical advice/attention.
First-aid measures after eye contact	: Rinse eyes with water as a precaution. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention.
First-aid measures after ingestion	: Rinse mouth and spit the fluids out. If the person is fully conscious, make him/her drink water (8 ounces / 240mL). Never give an unconscious person anything to drink. Do NOT induce vomiting. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. If you feel unwell, seek medical advice.

4.2. Most important symptoms/effects, acute and delayed

Symptoms/effects after inhalation	: May cause respiratory irritation.
Symptoms/effects after skin contact	: Repeated or prolonged contact may cause slight irritation to the skin.
Symptoms/effects after eye contact	: May cause slight irritation.
Symptoms/effects after ingestion	: Ingestion may cause nausea and vomiting.
Most Important Symptoms/Effects	: To our knowledge, the product does not present any particular risk, under normal conditions of use.

4.3. Indication of immediate medical attention and special treatment needed, if necessary

Other medical advice or treatment	: Treat symptomatically.
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SECTION 5: Fire-fighting measures

5.1. Suitable (and unsuitable) extinguishing media

Suitable extinguishing media	: Dry chemical, CO2, or water spray or regular foam.
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According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Unsuitable extinguishing media : Do not use a heavy water stream.

5.2. Specific hazards arising from the chemical

Fire hazard : Combustible liquid.
Hazardous decomposition products in case of fire : Toxic fumes may be released. Carbon dioxide. Carbon monoxide. Nitrogen oxides. Sulfur oxides.

5.3. Special protective equipment and precautions for fire-fighters

Firefighting instructions : Fight fire from safe distance and protected location. Do not enter fire area without proper protective equipment, including respiratory protection. Use extinguishing media appropriate for surrounding fire. Move containers from fire area if it can be done without personal risk.
Protection during firefighting : Do not attempt to take action without suitable protective equipment. Self-contained breathing apparatus. Complete protective clothing.

SECTION 6 Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

General measures : Evacuate area. Do not take actions involving personal risks. Avoid all personal contact including breathing in gas, vapors, mist, and spray. Stop leak if safe to do so. Absorb spillage to prevent material-damage. Notify authorities if product enters sewers or public waters.

For non-emergency personnel

Protective equipment : Wear recommended personal protective equipment.
Emergency procedures : Evacuate the danger area. If outdoors, move to an area upwind of the danger area. Avoid contact with skin and eyes. If possible without taking personal risks, Remove ignition sources, ventilate area. Prevent other non-emergency personnel from entering the danger area. No open flames, no sparks, and no smoking.

For emergency responders

Protective equipment : Wear the recommended personal protective equipment.
Emergency procedures : Evacuate personnel to a safe area. Ventilate spillage area. Stop leak if safe to do so.

Environmental precautions : Follow product instructions and best practices for use to avoid misapplications of the product. Do not negligently allow product to enter off-target drains, sewers, or ground water.

6.2. Methods and materials for containment and cleaning up

For containment : Contain with non-combustible inert absorbent.
Methods for cleaning up : Take up in non-combustible inert absorbent and place into container for disposal. Contaminated absorbent material may pose the same hazard as the spilt product. Decontaminate surfaces and equipment with water and detergent. Until a sufficient level of dilution is achieved, the decontamination water may pose the same hazards as the product.
Other information : Dispose of materials or solid residues at an authorized site.

For further information refer to section 8: "Exposure controls/personal protection", For further information refer to section 13

SECTION 7 Handling and storage

7.1. Precautions for safe handling

Precautions for safe handling : Do not handle until all safety precautions have been read and understood. Use only outdoors or in a well-ventilated area. Wear personal protective equipment. Do not get in eyes, on skin, or on clothing. Do not breathe mist, spray, vapors, gas.
Hygiene measures : Do not eat, drink or smoke when using this product. Always wash hands after handling the product. Contaminated work clothing should not be allowed out of the workplace.

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According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Additional hazards when processed : Proper grounding procedures to avoid static electricity should be followed.

7.2. Conditions for safe storage, including incompatibilities

Storage conditions : Store in a cool, dry and well-ventilated area away from incompatible substances. Protect from sunlight. Keep container closed when not in use.

Heat-ignition : Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

SECTION 8 Exposure controls/personal protection

8.1. Control parameters

Propylene glycol (57-55-6)

USA - AIHA - Occupational Exposure Limits

WEEL TWA	10 mg/m ³ aerosol
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8.2. Appropriate engineering controls

Appropriate engineering controls : Ensure good ventilation of the work station. Use general ventilation, local exhaust ventilation, or process enclosure to keep the airborne concentrations below the permissible exposure limits. Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure.

8.3. Individual protection measures, such as personal protective equipment

Personal protective equipment:

Personal protective equipment should be chosen according to national standards and in discussion with the supplier of the protective equipment. Wear recommended personal protective equipment.

Hand protection:

Wear protective gloves. The following materials are suitable for protective gloves: Butyl rubber, Nitrile rubber

Eye protection:

Chemical goggles. Safety glasses with side guards should be worn to prevent injury from airborne particles and/or other eye contact with this product

Skin and body protection:

Wear suitable protective clothing

Respiratory protection:

If vapors or mists exceed acceptable levels, wear NIOSH approved air-purifying respirator with cartridges/canisters approved for use against pesticides.

Personal protective equipment symbol(s):



SECTION 9 Physical and chemical properties

9.1. Basic physical and chemical properties

Physical state : Liquid

Appearance : Clear liquid.

Color : Yellowish

Odor : Sweet Floral

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According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Odor threshold	: No data available
pH	: 5.19 – 6.19
Melting point	: No data available
Freezing point	: No data available
Boiling point	: No data available
Flash point	: No data available
Flammability (solid, gas)	: Not applicable.
Vapor pressure	: No data available
Relative vapor density at 20°C	: No data available
Relative density	: No data available
Solubility	: No data available
Partition coefficient n-octanol/water (Log Pow)	: No data available
Auto-ignition temperature	: No data available
Decomposition temperature	: No data available
Viscosity, kinematic	: No data available
Explosion limits	: No data available
Particle characteristics	: No data available

9.2. Data relevant with regard to physical hazard classes (supplemental)

No additional information available

SECTION 10 Stability and reactivity

10.1. Reactivity

Combustible liquid.

10.2. Chemical stability

Stable under normal conditions of use.

10.3. Possibility of hazardous reactions

No dangerous reactions known under normal conditions of use.

10.4. Conditions to avoid

Incompatible materials.

10.5. Incompatible materials

Strong acids. Strong bases. Oxidizing agents.

10.6. Hazardous decomposition products

Thermal decomposition generates: Carbon dioxide. Carbon monoxide. Nitrogen oxides. Sulfur oxides.

SECTION 11 Toxicological information

11.1. Information on toxicological effects

Acute toxicity (oral)	: Not classified
Acute toxicity (dermal)	: Not classified
Acute toxicity (inhalation)	: Not classified

Curator	
LD50 oral rat	> 5000 mg/kg body weight
LD50 dermal rat	> 2000 mg/kg body weight

Curator

Safety Data Sheet

According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Skin corrosion/irritation : Slightly irritating to rabbits on cutaneous application
pH: 5.19 – 6.19

Serious eye damage/irritation : Slightly irritant but not relevant for classification
pH: 5.19 – 6.19

Respiratory or skin sensitization : Skin sensitization: Not classified.

Curator	
Additional information	Not a dermal sensitizer in mice by local lymph node assay (LLNA)
Germ cell mutagenicity	: Not classified
Carcinogenicity	: Not classified
Reproductive toxicity	: Not classified
STOT-single exposure	: Not classified
STOT-repeated exposure	: Not classified
Aspiration hazard	: Not classified
Symptoms/effects after inhalation	: May cause respiratory irritation.
Symptoms/effects after skin contact	: Repeated or prolonged contact may cause slight irritation to the skin.
Symptoms/effects after eye contact	: May cause slight irritation.
Symptoms/effects after ingestion	: Ingestion may cause nausea and vomiting.
Most Important Symptoms/Effects	: To our knowledge, the product does not present any particular risk, under normal conditions of use.

SECTION 12 Ecological information

12.1. Ecotoxicity

Ecology - general : The product is not considered harmful to aquatic organisms or to cause long-term adverse effects in the environment.

Hazardous to the aquatic environment, short-term (acute) : Not classified

Hazardous to the aquatic environment, long-term (chronic) : Not classified

Propylene glycol	
EC50 - Crustacea [1]	1000 mg/l
ErC50 algae	1000 mg/l
NOEC chronic crustacea	1000 mg/l
NOEC chronic algae	1000 mg/l

12.2. Persistence and degradability

Curator	
Persistence and degradability	Not established.
Propylene glycol	
Persistence and degradability	Rapidly degradable
Mefenoxam	
Persistence and degradability	Not rapidly degradable

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Safety Data Sheet

According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Propamacarb HCL

Persistence and degradability	Not rapidly degradable
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12.3. Bioaccumulative potential

No additional information available

12.4. Mobility in soil

No additional information available

12.5. Other adverse effects

Ozone	: Not classified
Fluorinated greenhouse gases	: No

SECTION 13 Disposal considerations

Regional waste regulation	: Disposal must be done according to official regulations.
Waste treatment methods	: Dispose of contents/container in accordance with licensed collector's sorting instructions.
Sewage disposal recommendations	: Disposal must be done according to official regulations.
Product/Packaging disposal recommendations	: Disposal must be done according to official regulations. Dispose of this material and its container at hazardous or special waste collection point. Refer to all applicable national, international and local regulations or provisions.

SECTION 14 Transport information

In accordance with DOT / IMDG / IATA

DOT	IMDG	IATA
14.1. UN number		
NA1993	Not regulated	Not regulated
14.2. Proper Shipping Name		
Combustible liquid, n.o.s. (Contains Methanol)	Not regulated	Not regulated
14.3. Transport hazard class(es)		
Combustible liquid	Not regulated	Not regulated
14.4. Packing group		
III	Not regulated	Not regulated
14.5. Environmental hazards		
	Not regulated	
No supplementary information available		

14.6. Transport in bulk

Not applicable

14.7. Special precautions for user

DOT	
UN-No. (DOT)	: NA1993
DOT Packaging Exceptions (49 CFR 173.xxx)	: 150
DOT Packaging Non Bulk (49 CFR 173.xxx)	: 203
DOT Packaging Bulk (49 CFR 173.xxx)	: 241

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According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

DOT Quantity Limitations Passenger aircraft/rail (49 : 60 L

CFR 173.27)

DOT Quantity Limitations Cargo aircraft only (49 : 220 L

CFR 175.75)

DOT Vessel Stowage Location : A - The material may be stowed "on deck" or "under deck" on a cargo vessel and on a passenger vessel.

IMDG

Not regulated

IATA

Not regulated

SECTION 15 Regulatory information

15.1. Federal regulations

All components of this product are present and listed as Active on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory, except for:

Mefenoxam	CAS-No. 70630-17-0	4%
Propamacarb HCL	CAS-No.	40%
Trade Secret	CAS-No. Trade Secret	Trade Secret%

This product or mixture is not known to contain a toxic chemical or chemicals in excess of the applicable de minimis concentration as specified in 40 CFR §372.38(a) subject to the reporting requirements of section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 and 40 CFR Part 372.

FIFRA Labelling

EPA Registration Number	228-771
This chemical is a pesticide product registered by the U.S. Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. The following is the hazard information as required on the pesticide label.	
FIFRA Signal Word	Caution
FIFRA Precautionary Statement	Keep out of reach of children. Harmful if absorbed through skin. Causes moderate eye irritation.
FIFRA Human Health Hazards	Harmful if absorbed through skin. Causes moderate eye irritation. Avoid contact with skin, eyes or clothing. Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco or using the toilet. Remove and wash contaminated clothing before reuse.
FIFRA First Aid	IF ON SKIN OR CLOTHING • Take off contaminated clothing. • Rinse skin immediately with plenty of water for 15-20 minutes. • Call poison control center or doctor for further treatment advice IF IN EYES • Hold eye open and rinse slowly and gently with water for 15-20 minutes. • Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eyes. • Call a poison control center or doctor for treatment advice.

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Safety Data Sheet

According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

15.2. International regulations

CANADA

Propylene glycol (57-55-6)
Listed on the Canadian DSL (Domestic Substances List)
Mefenoxam (70630-17-0)
Not listed on the Canadian DSL (Domestic Substances List)/NDSL (Non-Domestic Substances List)
Propamacarb HCL
Not listed on the Canadian DSL (Domestic Substances List)/NDSL (Non-Domestic Substances List)

EU-Regulations

No additional information available

National regulations

Propylene glycol (57-55-6)
Listed on INSQ (Mexican National Inventory of Chemical Substances)
Mefenoxam (70630-17-0)
Listed on INSQ (Mexican National Inventory of Chemical Substances)
Propamacarb HCL
Listed on INSQ (Mexican National Inventory of Chemical Substances)

15.3. State regulations

 **WARNING:** This product can expose you to Methanol, which is known to the State of California to cause birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

Component	State or local regulations
Propylene glycol(57-55-6)	U.S. - New Jersey - Right to Know Hazardous Substance List; U.S. - Pennsylvania - RTK (Right to Know) List

SECTION 16 Other information

According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Revision date : 11/4/2025
Issue date : 10/21/2024

Full text of hazard classes and H-statements	
H227	Combustible liquid
H302	Harmful if swallowed
H315	Causes skin irritation
H317	May cause an allergic skin reaction
H318	Causes serious eye damage

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Safety Data Sheet

According to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Indication of changes:		
Section	Changed item	Comments
3	Composition/Information on ingredients	Modified
9	Appearance	Added
9	Color	Modified
11	Serious eye damage/irritation - Comment	Modified
15	FIFRA First Aid	Added
15	FIFRA Human Health Hazards	Added
15	FIFRA Precautionary Statement	Added
15	FIFRA Signal Word	Added
15	EPA Registration Number	Modified

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.